

Business Requirements Document (BRD)

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Project Name: Autonomous Delivery Cart

Statement of Confidentiality

The information contained in this document and related artifacts constitute confidential information of Autonomous Delivery Cart and are intended for internal usage purposes only.

Document Control

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How to read this document?

The document has been structured into the following sections:

1. Purpose

This section will provide the purpose of this document in terms of CARTA's reporting requirements.

2. Overview

This section will provide an overview and the background of the project. It also highlights the project goal, in-scope project activities, project dependencies and the stakeholders. The entire project cycle has been divided into activities.

3. Key Assumptions

This section will provide the list of project assumptions considered to deliver the project scope.

4. Business Requirements

This section will provide the list of business requirements, High level process flow with key business terms and definitions and Key Measure

1. Purpose

The purpose of this Business Requirements Document (BRD) is to provide a detailed description of the business needs and functional requirements for the Autonomous Delivery Cart System. This document will serve as the primary agreement between the project team and its supervisors, ensuring a shared understanding of the project's goals, scope, and success criteria.

2. Overview

2.1 Project Overview and Background

Manual, last-mile delivery within closed environments (e.g., residential compounds, hotels) is often inefficient, costly, and lacks real-time visibility. The Autonomous Delivery Cart System aims to solve this problem by introducing a smart, automated solution.

This project will deliver a proof-of-concept system comprising a cloud backend, a customer-facing mobile application, a vendor web portal, and an IoT-enabled autonomous cart. The system will provide a faster, contactless, and more reliable delivery experience, laying the groundwork for a potential future start-up.

2.2 Benefits for Business

1. Increased Operational Efficiency

This is about making the delivery process cheaper, faster, and more reliable.

- **Reduced Labor Costs:** The most direct benefit. By automating deliveries, the system reduces the need for dedicated delivery staff, saving significant costs on salaries and associated expenses.
- **Faster Delivery Times:** Autonomous carts can operate on optimized paths without breaks, leading to quicker fulfillment and higher customer satisfaction.
- **Higher Throughput:** A single vendor can manage multiple outgoing orders simultaneously without being limited by the number of available delivery drivers. This allows them to serve more customers during peak hours.

2. Enhanced Customer Experience & Satisfaction

This focuses on making the service more appealing and convenient for the end-user.

- **On-Demand Convenience:** Customers are no longer tied to a store's limited delivery schedule. They can order what they need, when they need it, and receive it quickly.
- **Real-Time Transparency:** The ability to track the cart's location in real-time removes uncertainty and anxiety for the customer, providing a modern and professional experience.
- **Contactless & Secure Delivery:** The secure OTP unlock mechanism provides a safe,

contactless pickup process, which is a major selling point for health-conscious consumers.

3. Data-Driven Decision Making

This is about leveraging the data your system collects to make smarter business choices. Your BRD's KPI section is perfect for this.

- **Actionable Insights:** The Manager Dashboard provides immediate access to crucial metrics like Order Completion Rate and Cart Utilization. A manager can instantly see if a process is inefficient or if a cart requires maintenance, preventing problems before they impact customers.
- **Understanding Customer Behavior:** By analyzing Customer Order Frequency and popular products, vendors can optimize their inventory, run targeted promotions, and better understand the needs of the community they serve.
- **Performance Monitoring:** The system allows for clear monitoring of vendor performance, ensuring a consistently high quality of service across the entire platform.

4. Future-Proofing and Scalability

This highlights the long-term strategic value of building this system.

- **Foundation for Growth:** Your project isn't just a delivery system; it's a scalable logistics platform. It's designed to grow from one cart to a fleet, and from one compound to many, with minimal changes to the core architecture.
- **Adaptability:** The system is built to evolve. Integrating new features like payment gateways, advanced AI for route optimization, or different types of delivery carts becomes much easier with this solid foundation in place.
- **Competitive Advantage:** For any hotel, compound, or university, offering an exclusive, automated delivery service is a significant and modern amenity that sets them apart from competitors.

2.3 Project Dependencies

The successful implementation of this project is dependent on the following:

- **AWS Cloud Services:** Consistent access to and funding for required AWS services (IoT Core, Lambda, RDS, etc.).
- **Hardware Availability:** Access to the specified IoT hardware (Jetson Nano, sensors, cart chassis).
- **Network Connectivity:** The operational environment (e.g., a university campus) must have stable Wi-Fi or 4G/5G coverage for the carts.

2.4 Stakeholders

Stakeholder	Role	Interest in Project
Customers	End-Users	Place orders, track deliveries, and receive goods. Value convenience and speed.

Vendors	Business Users	Fulfill orders, manage inventory, and dispatch carts. Value efficiency and order management.
Managers	System Administrators	Monitor system health, view analytics, and manage users. Value operational oversight and data insights.
Supervisors	Academic Evaluators	Assess the project's technical complexity, completeness, and success in meeting its defined goals.

3. Key Assumptions

- All users (customers, vendors, managers) will have access to a modern smart device (phone or computer) and internet connectivity.
- The initial MVP will not include a payment gateway; all orders are assumed to be "cash on delivery" or pre-paid.
- The cart's autonomous navigation will operate on predefined, safe pathways within the test environment.

4. Business Requirements

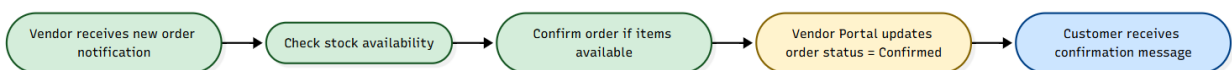
4.1 Business Process Flow

The core business process for a delivery is as follows:

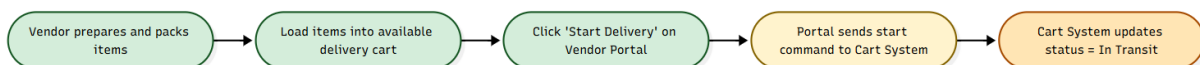
1. **Order Placement:** A registered customer uses the mobile app to browse products from a specific vendor and places an order for delivery.



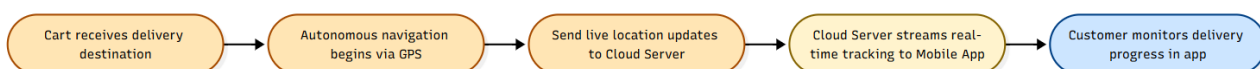
2. **Order Confirmation:** The vendor receives the new order on their web portal, confirms they have the items in stock, and begins preparing the order.



3. **Cart Dispatch:** The vendor loads the prepared items into an available cart and triggers the "Start Delivery" command from their portal.



4. **Autonomous Delivery:** The cart receives the command and autonomously navigates to the customer's specified delivery location. The customer can track the cart's progress in real-time on the mobile app.



5. **Customer Verification & Pickup:** Upon arrival, the system sends an SMS with a One-Time Password (OTP) to the customer. The customer enters the OTP into the app, which securely unlocks the cart's storage compartment.



6. **Return to Base:** After the customer retrieves their items and closes the door, the cart automatically begins its journey back to the vendor's location.



4.2 Functional Business Rules

This section outlines the specific rules the system must enforce.

User & Authentication:

- **BR-01:** All users must be authenticated to access system functions.
- **BR-02:** User accounts (especially email and phone) must be unique.
- **BR-03:** The system must enforce distinct permissions for Customer, Vendor, and Manager roles.

Order Management:

- **BR-04:** Orders must be associated with a single customer and a single vendor.
- **BR-05:** An order's status must follow a logical lifecycle (pending -> preparing -> delivering -> completed).
- **BR-06:** The price of items is locked at the time of purchase and stored with the order.

Cart & Delivery:

- **BR-07:** The cart's door can only be unlocked via a successful customer OTP verification.
- **BR-08:** The OTP code must expire after 5 minutes to ensure security.
- **BR-09:** A cart must return to its base automatically after a completed delivery.

System & Security:

- **BR-10:** A manager must have an "emergency stop" capability for any active cart.
- **BR-11:** All critical administrative actions (e.g., deactivating a vendor) must be recorded in an audit log.
- **BR-12:** All communication channels must be encrypted (TLS/SSL).

4.3 Defined Key KPIs/Metrics/Measures

The success and performance of the system will be measured by the following Key Performance Indicators (KPIs), which will be visible on the Manager Dashboard.

KPI / Metric	Description	Business Question Answered
Order Completion Rate	The percentage of orders created that are successfully completed.	How efficient is our delivery process?
Average Delivery Time	The average time from when a vendor dispatches a cart to when the customer unlocks it.	How fast are our deliveries?
Cart Utilization Rate	The percentage of time a cart is actively delivering versus being idle or charging.	Are we making efficient use of our hardware assets?
Order Accuracy	(Future KPI) The rate of orders completed without any reported issues (e.g., wrong items).	Are vendors fulfilling orders correctly?
Customer Order Frequency	The average number of orders a customer places per month.	Are our customers engaged and returning?
System Uptime	The percentage of time the cloud backend and APIs are operational and available.	Is our system reliable?

4.4 Business Terms and Definitions

This section provides definitions for key business and technical terms used throughout this document to ensure a common understanding among all stakeholders.

<i>Term</i>	<i>Definition</i>
<i>Autonomous Cart</i>	<i>The IoT-enabled, self-driving robotic unit responsible for transporting goods from the vendor to the customer.</i>
<i>Cloud Backend</i>	<i>The server-side infrastructure and logic hosted on AWS that manages all system operations, including orders, users, and cart communication.</i>
<i>IoT (Internet of Things)</i>	<i>The network of physical devices (the delivery carts) embedded with sensors and software that allows them to connect and exchange data with the cloud.</i>
<i>Key Performance Indicator (KPI)</i>	<i>A quantifiable measure used to evaluate the system's performance against its business objectives (e.g., Average Delivery Time).</i>
<i>Minimum Viable Product (MVP)</i>	<i>The initial version of the system that includes just enough core features to be functional and demonstrate the end-to-end delivery cycle.</i>
<i>One-Time Password (OTP)</i>	<i>A unique, time-sensitive code sent to a customer's phone via SMS to securely verify their identity and authorize the unlocking of the cart's door.</i>
<i>Stakeholder</i>	<i>Any individual or group with an interest in the project's outcome, including Customers, Vendors, Managers, and Supervisors.</i>
<i>Vendor</i>	<i>A business (e.g., supermarket, pharmacy) that lists its products on the platform and uses the system to fulfill customer orders.</i>

5. Approvals & Acknowledgements

The undersigned approve the business requirements outlined in this document.

Role	Name	Signature	Date